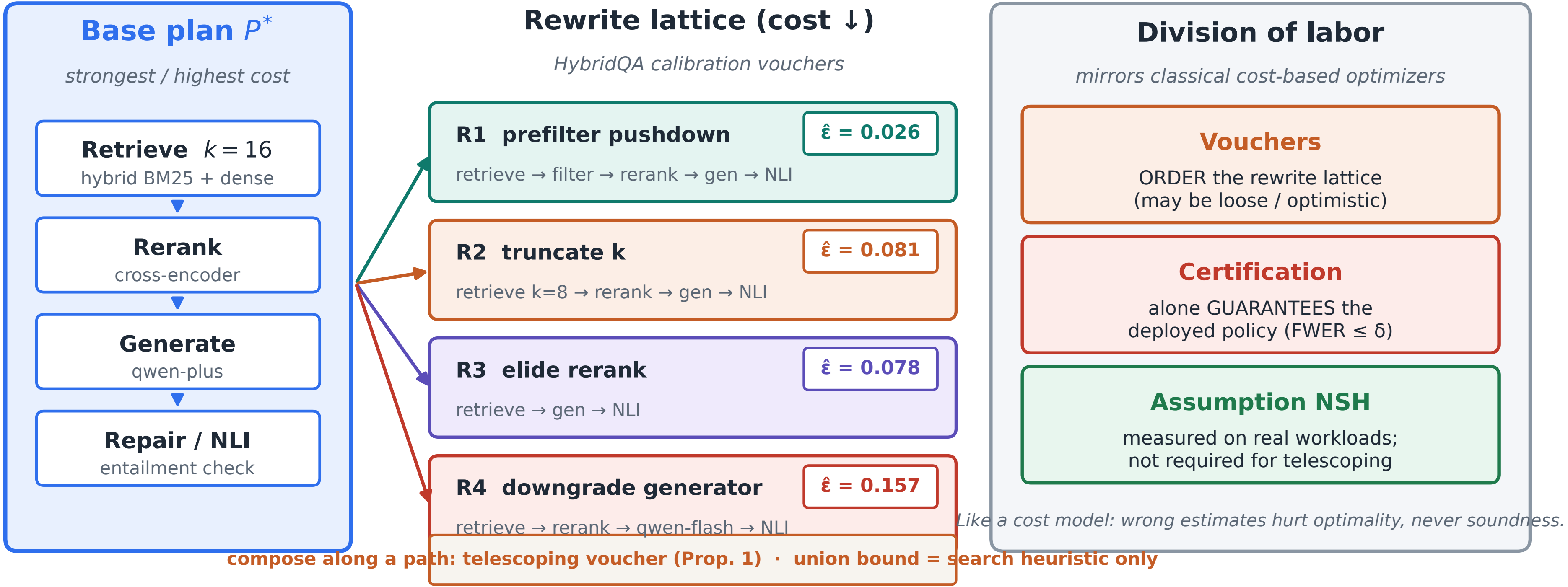
Base plan P^*
strongest / highest cost
Retrieve $k = 16$
hybrid BM25 + dense
Rerank
cross-encoder
Generate
qwen-plus
Repair / NLI
entailment check
Rewrite lattice (cost ↓)
HybridQA calibration vouchers
R1 prefilter pushdown
$\hat{\varepsilon}$ = 0.026
retrieve → filter → rerank → gen → NLI
R2 truncate k
$\hat{\varepsilon}$ = 0.081
retrieve k=8 → rerank → gen → NLI
R3 elide rerank
$\hat{\varepsilon}$ = 0.078
retrieve → gen → NLI
R4 downgrade generator
$\hat{\varepsilon}$ = 0.157
retrieve → rerank → qwen-flash → NLI
compose along a path: telescoping voucher (Prop. 1) · union bound = search heuristic only
Division of labor
mirrors classical cost-based optimizers
Vouchers
ORDER the rewrite lattice
(may be loose / optimistic)
Certification
alone GUARANTEES the
deployed policy (FWER ≤ δ)
Assumption NSH
measured on real workloads;
not required for telescoping
Like a cost model: wrong estimates hurt optimality, never soundness.